

# Aman Kumar Sahu

+91 9664866438 ◊ [Github](#) ◊ [Linkedin](#) ◊ [sahuaman454@gmail.com](mailto:sahuaman454@gmail.com)

I am a passionate Data Scientist with over 3+ years of experience building ML and Gen AI solutions, with core expertise in LLM pipelines, GraphRAG, Advanced RAGs, Text-to-SQL., Agentic Systems, Finetuning LLMs and knowledge graph-driven automation for real-world enterprise use cases.

## TECHNIAL STRENGTHS

|                       |                                                                                                                      |
|-----------------------|----------------------------------------------------------------------------------------------------------------------|
| Languages             | Python, SQL, C, C++, Scala (Beginner)                                                                                |
| Technologies          | Machine Learning, LLM, AI and Deep Learning, Gen-AI, ChatGPT, RAG, GraphRAG, Vector DB, Retrievers,MCP               |
| Framework/Libraries   | Pandas, NumPy, Scikit, TensorFlow, Keras, Tableau, Langchain, LangGraph, Langsmith, Dash Leaflet, Plotly Dash, RAGAs |
| Vector Store/VectorDB | FAISS, ChromaDB , Neo4j, Pinecone, OpenSearch, Azure AI Search                                                       |

## WORK EXPERIENCE

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| KPMG Global Services- Data Scientist Consultant - Gen AI (Full time)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | Sept 2025 - Present    |
| <ul style="list-style-type: none"><li><b>AI-Powered Transaction Monitoring (RAG):</b> Built an AI-driven transaction monitoring POC for <b>procurement and Accounts Payable (AP) controls</b>, using <b>RAG-based extraction</b> to standardize invoice, PO, and contract data. Implemented a <b>multi-stage matching framework</b> (exact, fuzzy, TF-IDF, embeddings) to reconcile invoice line items against contract terms, enabling <b>explainable detection of pricing variances, contract non-compliance, and missed discounts</b> with <b>audit-ready outputs</b>.</li><li><b>Enterprise Contract Intelligence (CRAG):</b> Built a CRAG-driven system using <b>Azure OpenAI &amp; AI Search</b> to extract <b>20 compliance-critical fields</b> from <b>2,400+ contracts</b> with <b>&gt;90% accuracy</b>. Designed four retrieval indexes with <b>hybrid</b> (keyword + semantic search), field-level retrieval validation, and source traceability for audit-ready, compliance-aligned outputs. Developed a <b>135-record annotated dataset</b> and applied <b>targeted upsampling for sparse fields</b>, maintaining GitHub <b>prompt/model versioning</b> for reproducibility.</li><li><b>Technology Stack:</b> <i>Python, NumPy, NLTK, Scikit-learn, Seaborn, Plotly and Matplotlib, Gen-AI, Langraph, Langchain, Prompt Engineering, vector stores, RAG, Azure OpenAI, Azure AI Search , Azure, AWS Sagemaker,RAGAs, MCP, AI Agents</i></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                        |
| Arcadis - Data Scientist Associate Consultant (Full time)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | June 2023 – Aug 2025   |
| <ul style="list-style-type: none"><li><b>Tap Card Digitization:</b> Led the end-to-end digitization of <b>10.6K+ Tap Cards</b>, building a template-classification pipeline and training <b>Azure Custom Document Extraction models</b> to extract key fields (e.g., SVC pipe size, date), delivering structured outputs for QC review.</li><li><b>GraphRAG using Neo4j:</b> Automated structured field extraction from unstructured legal PDFs by combining AzureOCR with LangChain’s LLMGraphTransformer to build a Neo4j-based knowledge graph, enabling Cypher querying and LLM-powered Q&amp;A over graph data.</li><li><b>Text to SQL (Phase 2):</b> Enhanced the system by introducing LangChain’s <b>SQLToolkit</b> with <b>ReAct</b> Agent and migrating the architecture using LangGraph, enabling multi-turn query handling and Human in the Loop.</li><li><b>Text to SQL (Phase 1):</b> Developed a Text to SQL system using Streamlit and OpenAI – <b>GPT – 4o</b>, reducing data retrieval times by 50% and enhancing accessibility for non-technical users.</li><li><b>Retrieval-Augmented Generation (RAG) System:</b> Deployed a RAG system using LangChain, FAISS, <b>Google Flan T5</b> and Streamlit, enhancing data retrieval accuracy by 40% and completing the project within a month.</li><li><b>NRW Leak Detection:</b> Developed a GNN-based (<b>ChebNET Model</b>) leak detection system for WDNs, <b>achieving a 87% accuracy rate</b> in leak localization and reducing <b>water loss by 15%</b>.</li><li><b>IT Ticket Analysis:</b> Implemented advanced NLP techniques (keyBERT, Gensim, <b>Latent Dirichlet Allocation (LDA)</b>, GPT models) for efficient IT support, reducing ticket resolution time by <b>30% and improving service quality</b>.</li><li><b>Dynamic Spatial Analytics Web App:</b> Created a spatial data web app using Plotly Dash, Python, PostgreSQL and ArcGIS, improving spatial data utilization by <b>35%</b> and aiding decision-making processes.</li><li><b>Technology Stack:</b> <i>Python, NumPy, NLTK, Scikit-learn, TensorFlow, Keras, Seaborn, Plotly and Matplotlib, Gen-AI, Langraph, Langchain, Prompt Engineering, vector stores, RAG, llama, MS Fabric and PowerBI, Gradio, KNIME, Plotly Dash, Dash Leaflet,</i></li></ul> |                        |
| GEP Worldwide - Data Engineer / Data Scientist (Full time)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | June 2022 – June ,2023 |
| <ul style="list-style-type: none"><li>Enhanced data quality and efficiency, <b>reducing data errors by 30%</b> and <b>increasing processing speed by 20%</b>.</li><li>Conducted EDA, improved model <b>performance by 15%</b>, and optimized model accuracy.</li><li><b>Technology Stack:</b> <i>Azure Data Bricks, Scala/Python, Spark, Tableau, NumPy, Scikit, Pandas and Matplotlib.</i></li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                        |

## EDUCATION

|                                                                                     |            |
|-------------------------------------------------------------------------------------|------------|
| Diploma in Data Science - INDIAN INSTITUTE OF TECHNOLOGY, MADRAS                    | CGPI: 8.13 |
| Foundation Level (FY) - INDIAN INSTITUTE OF TECHNOLOGY, MADRAS                      | CGPI: 8.00 |
| B. Tech, Information Technology - 2018 - 2022 - K J SOMAIYA INSTITUTE OF TECHNOLOGY | CGPI: 9.04 |

## PROJECTS

|                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|
| Nano Satellite Project: BeliefSat-0 - Command and Telemetry                                                                                                                                                                                                                                                                                                                                                                                                 | July 2019 - June 2022 |
| <ul style="list-style-type: none"><li>Developed BeliefSat-0, a 2p-PocketQube nanosatellite, as part of K.J.S.I.T Satellite Group.</li><li>Promoted to Senior Command and Telemetry Developer for leadership and contributions.</li><li>Successfully launched <b>BeliefSat-0</b> aboard <b>ISRO's PSLV-C58 XPoSat</b> on January 1, 2024.</li><li>Enhanced team efficiency <b>approximately by 20%</b> and <b>reduced development time by 10%</b>.</li></ul> |                       |

## SOCIAL ACTIVITIES AND ACHIEVEMENTS

|                                                                                                      |
|------------------------------------------------------------------------------------------------------|
| Arcadis - Awarded with <b>2 Spot Bonuses</b> in a span of <b>four months (Nov 2024 and Feb 2025)</b> |
|------------------------------------------------------------------------------------------------------|